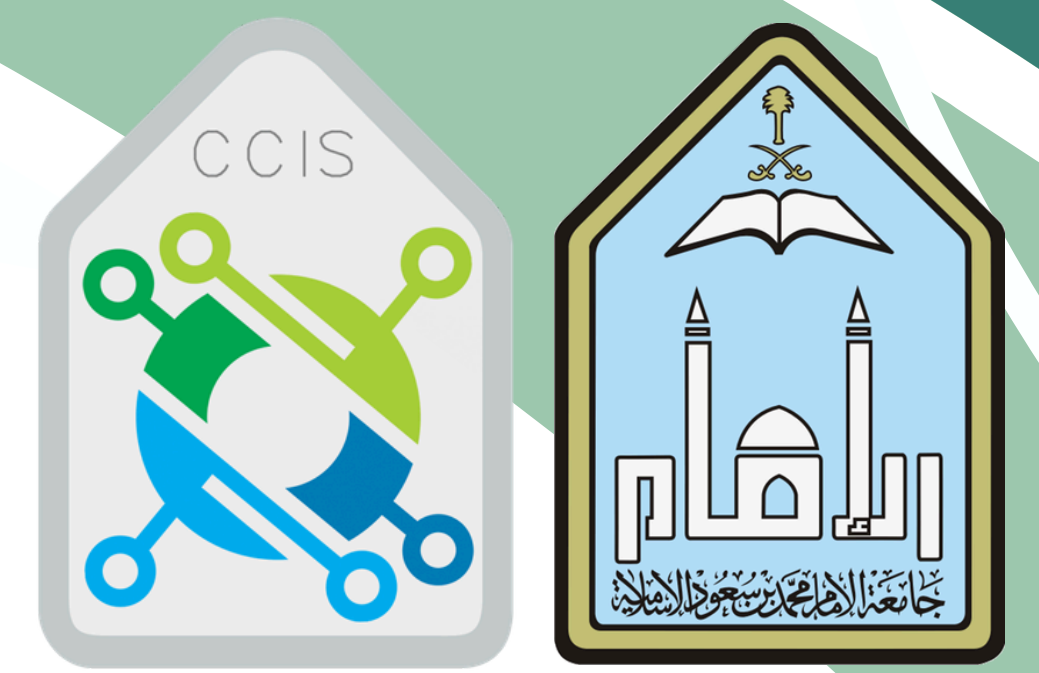




DRBK: AI-Powered Smart Cane for The Visually Impaired



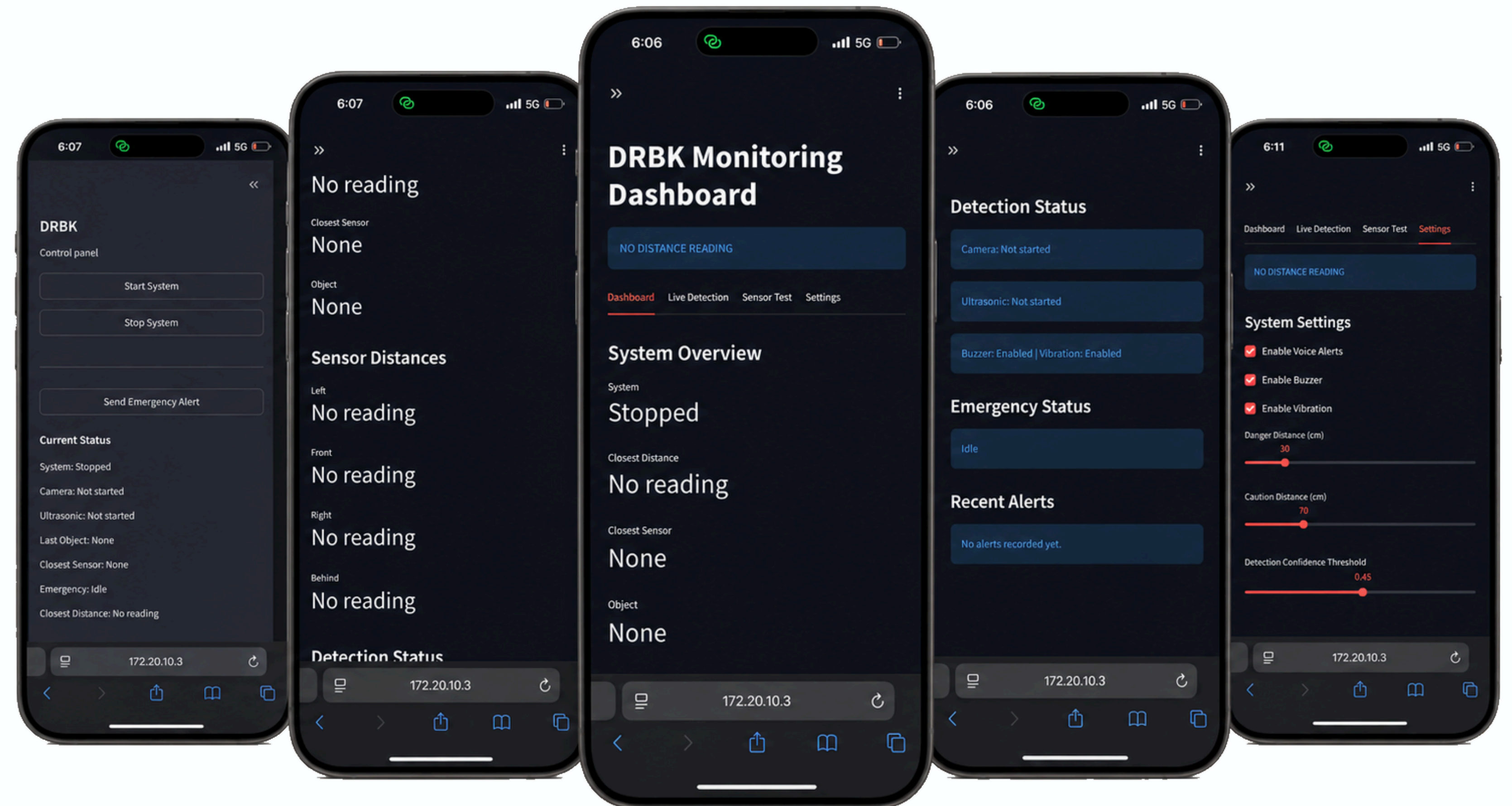
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Introduction

Safe independent mobility is one of the greatest challenges for visually impaired individuals in crowded streets, complex intersections, and unfamiliar indoor spaces. Traditional white canes detect only ground-level contact - they cannot identify obstacle types, recognize elevated hazards, or alert caregivers in emergencies. Recent advances in embedded sensing, edge AI, and IoT open the door to smarter assistive devices. DRBK bridges this gap by fusing ultrasonic distance sensing, camera-based object detection, and emergency communication into a single smart cane.

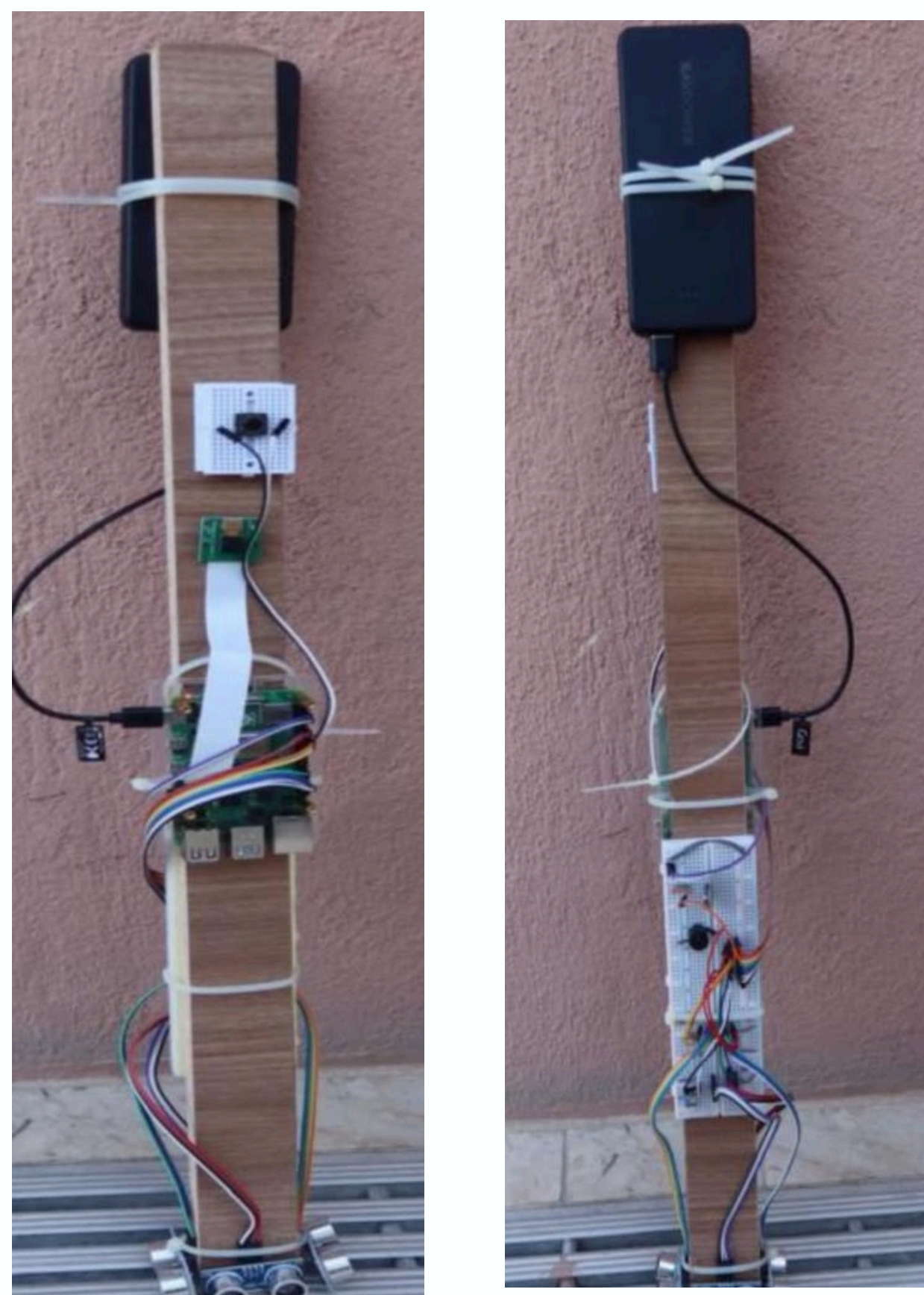
Website



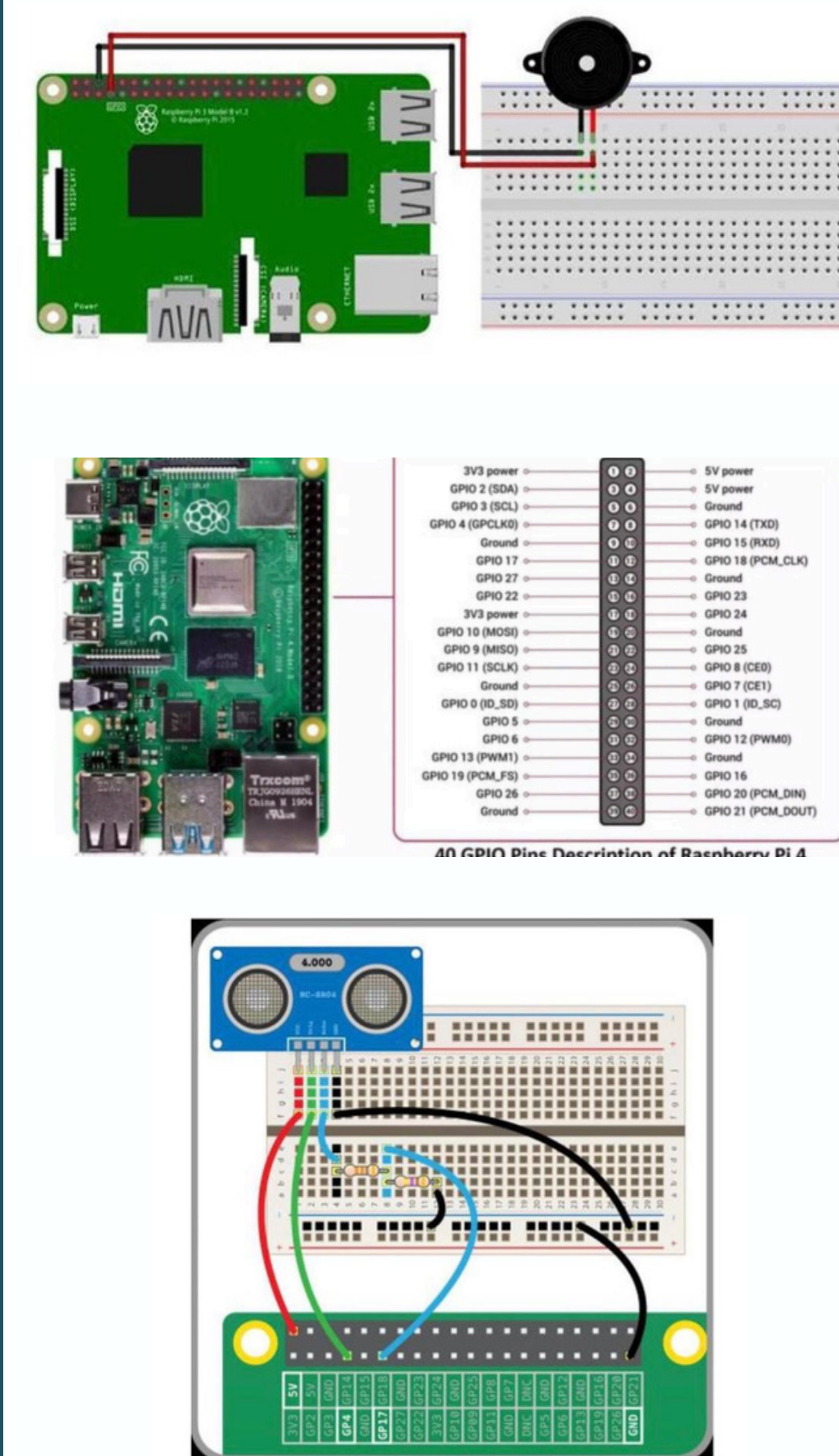
Objectives

- Detect obstacles in 4 directions using ultrasonic sensors.
- Classify nearby objects via camera + TFLite AI model.
- Deliver real-time voice, buzzer & vibration feedback.
- Send Telegram SOS alerts to caregiver on button press.
- Provide a Streamlit dashboard for monitoring & testing.
- Run efficiently on Raspberry Pi 4 edge hardware.

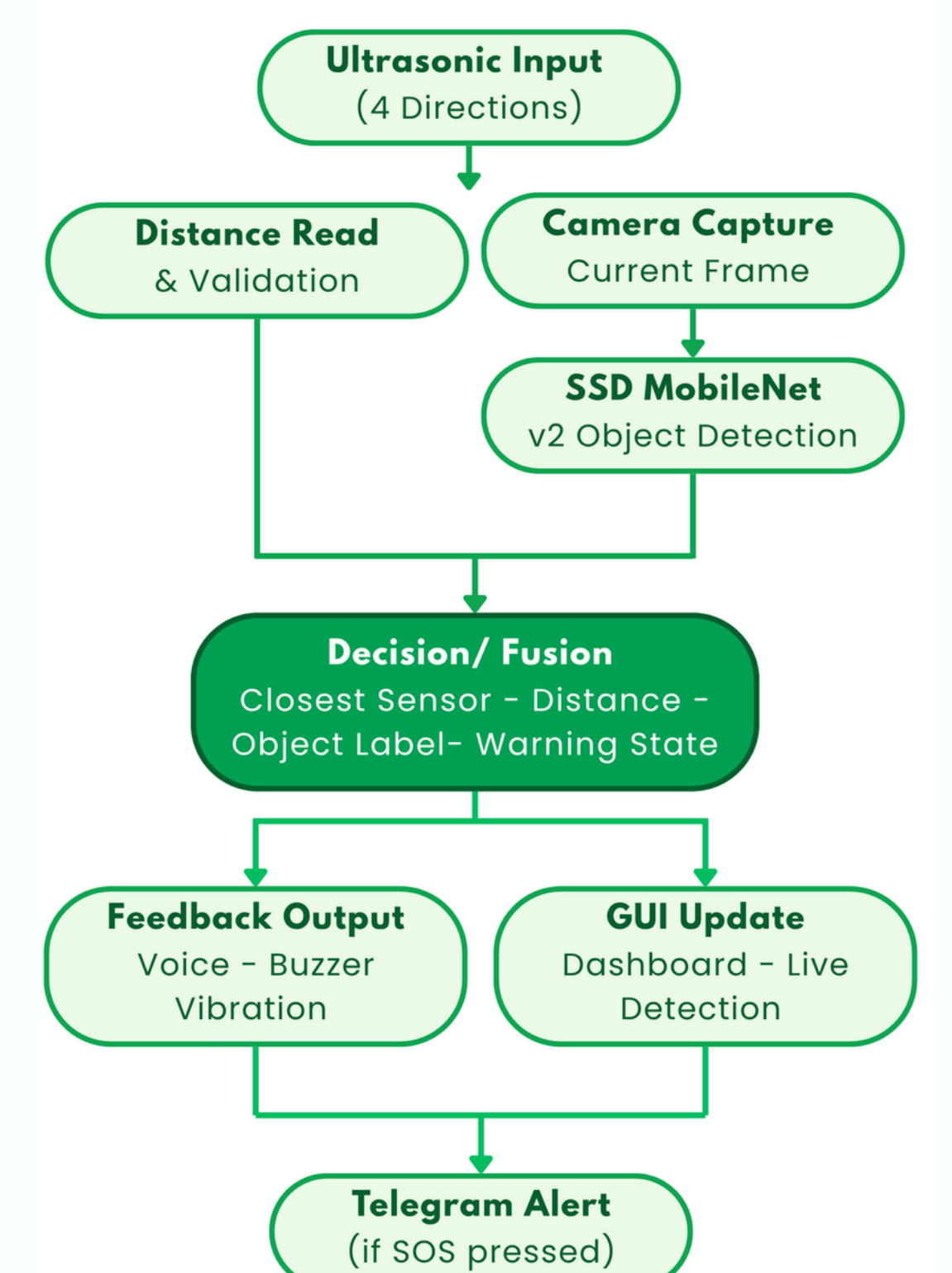
Prototype



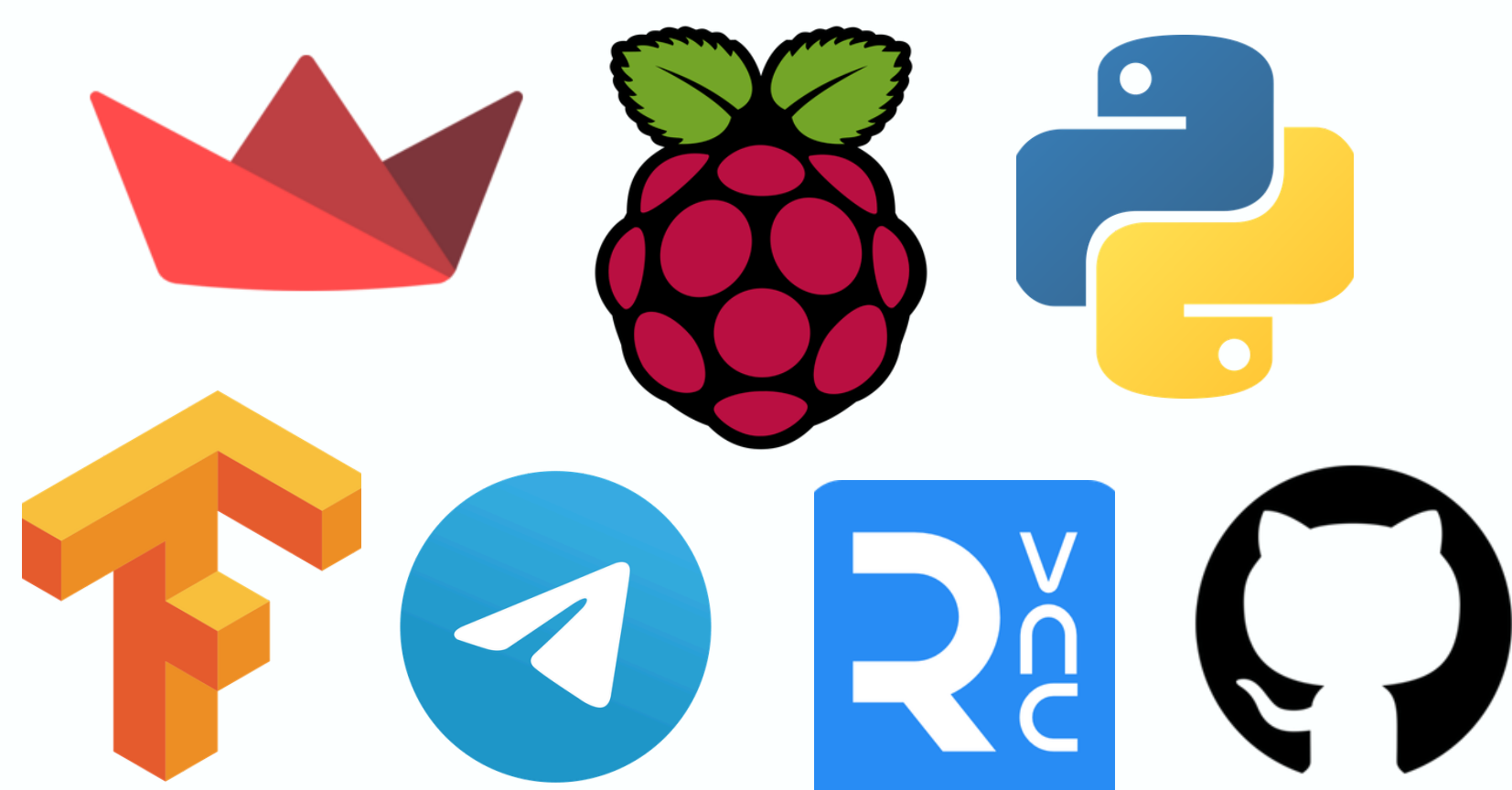
Components



Architecture



Tools



References



Methodology

DRBK was developed as an integrated AI-powered smart cane that combines ultrasonic sensors, camera-based object detection, and real-time feedback. Four ultrasonic sensors detect obstacle distance and direction, while a camera captures the scene for object recognition using TensorFlow Lite SSD MobileNet v2 on Raspberry Pi 4. The system classifies the situation into safe, caution, or danger and alerts the user through voice, vibration, and buzzer feedback. It also includes a Telegram SOS alert and a Streamlit GUI for monitoring and testing the system.

Conclusion & Future Work

DRBK proved the feasibility of using edge AI, IoT hardware, and computer vision to build a smart cane that supports visually impaired users through real-time obstacle detection, multi-directional sensing, object classification, feedback, and emergency alerts. Future improvements include enhancing detection accuracy, adding GPS voice guidance, developing a mobile app, integrating 3D mapping, testing with visually impaired users, and improving battery efficiency.

Results

4
Ultrasonic
Sensors

30cm
Danger
Threshold

0.20s
Refresh
Interval

SOS ✓
Telegram
Verified